

University of Cape Town

CSC2042S: Machine Learning

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1 Supervised learning

1.1 What is machine learning?

Definition: Machine Learning

Machine learning is the study of algorithms that:

- improve their performance P (according to a criterion that measures success)
- at some task T
- with experience E (from data examples).

A well-defined task is given by $< P, T, E >$. For example:

- **Image classification:** <Accuracy, Classify handwritten digits, Dataset of labelled images>.
- **Chess:** <Playing quality, Play chess against human, Dataset of old games>.
- **ChatGPT:** <Quality of conversation, Generate responses, All text on the web>.

1.2 Types of machine learning

There are many types:

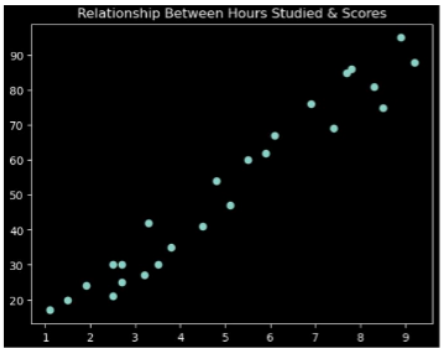
- Supervised learning.
- Unsupervised learning.
- Reinforcement learning.
- etc.

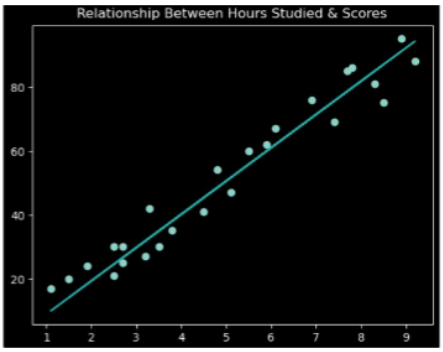
1.2.1 Unsupervised learning

The problem is framed as finding structure in x without labels (no y). Inputs are examples in the dataset represented by *features*. Data D is $|D|$ examples $\{x_1, x_2, \ldots, x_{|D|}\}$ without any labels. Given a new example x, where does this fit into the data?

1.3 Simple linear regression

A simple linear regression model predicts the output as a linear function of one input feature x: $y = wx + b$, otherwise known as a straight line function. Changes in x imply a proportional change in y. Even if the relationship is not perfectly linear, linear regression can still be effective as an approximation or baseline model.

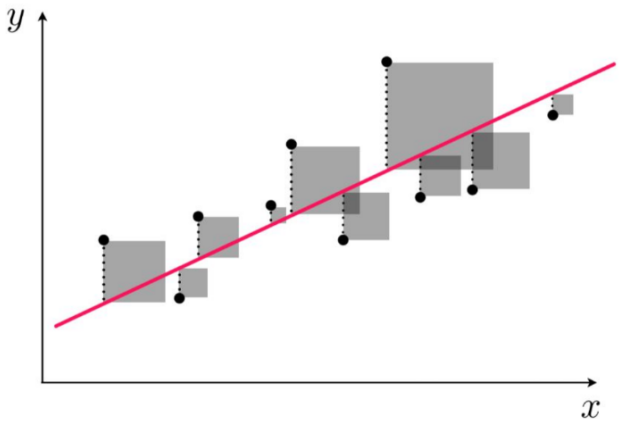




1.3.1 Loss function

For each input, linear regression produces an estimate $\hat{y}_i = w \cdot x_i + b$. We want to set w and b to minimize the distance between our estimate $\hat{y}_i$ (prediction) and the true y_i. We want to minimize this distance/error across our entire training set $\{(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)\}$.

A principled way to define the learning problem is with a loss function (also known as a cost function): $L(\hat{y}_i, y_i)$ measuring how close the model prediction $\hat{y}_i$ is to the observed data y_i. If the model fits the data well, so that $y_i \approx \hat{y}_i$, then the loss function should take a small value, and vice versa. For this, we can use the squared error loss.



The squared error loss measures how good your linear regression model with parameters w, b fits the data. For a single data point (x_i, y_i):

$$\begin{aligned} L(\hat{y}_i, y_i) &= (y_i - \hat{y}_i)^2 \\ &= (y_i - f(x_i; w, b))^2 \\ &= (y_i - (w \cdot x + b))^2 \\ &= (y_i - w \cdot x - b)^2 \end{aligned}$$

We want to find the weight that minimizes the loss, averaged over all examples in our training set.

$$\boxed{J(w, b) = \frac{1}{N} \sum_{i=1}^{N} L(\hat{y}_i, y_i)}$$

Note: Why as a function of w and b?

These are the variables we want to learn!

All examples (x_i, y_i) in the function are determined by whatever is in the training set. This is out of our control. Now, we want to find:

$$\hat{w}, \hat{b} = \underset{w,b}{\operatorname{argmin}} J(w, b)$$

For linear regression, the loss function (mean squared error) is convex. This means it has a single global minimum and no local minima. Linear regression has a closed form solution. The question is, how do we find $\hat{w}, \hat{b} = \underset{w,b}{\operatorname{argmin}} J(w, b)$?

1. Take the partial derivative of J with respect to w and b.
2. Set the partial derivative equal to 0.
3. Solve for w and b.

Derivation

Derivation w.r.t b:

$$\begin{aligned}
\frac{\partial}{\partial b} J(w, b) &= \frac{\partial}{\partial b} \frac{1}{N} \sum_{i=1}^{N} (y_i - wx_i - b)^2 \\
&= \frac{-2}{N} \sum_{i=1}^{N} (y_i - wx_i - b) \\
0 &= \sum_{i=1}^{N} (y_i - wx_i - b) \quad \text{(Setting equal to 0)} \\
0 &= \sum_{i=1}^{N} y_i - w \sum_{i=1}^{N} x_i - Nb \\
\hat{b} &= \bar{y} - \hat{w}\bar{x}
\end{aligned}$$

Derivation w.r.t w:

Before differentiating w.r.t w, substitute this expression for b back into the cost so that J becomes a function of w only:

$$J(w) = \frac{1}{N} \sum_{i=1}^{N} \Big[y_i - wx_i - (\bar{y} - w\bar{x}) \Big]^2 = \frac{1}{N} \sum_{i=1}^{N} \Big[(y_i - \bar{y}) - w(x_i - \bar{x}) \Big]^2$$

Now differentiate w.r.t w:

$$\frac{d}{dw} J(w) = \frac{-2}{N} \sum_{i=1}^{N} \big[(y_i - \bar{y}) - w(x_i - \bar{x}) \big] \big[(x_i - \bar{x}) \big]$$

Set the derivative to 0 and simplify:

$$0 = \frac{-2}{N}\Big[\sum_{i=1}^{N}(y_i - \bar{y})(x_i - \bar{x}) - w\sum_{i=1}^{N}(x_i - \bar{x})^2\Big]$$

$$w\sum_{i=1}^{N}(x_i - \bar{x})^2 = \sum_{i=1}^{N}(y_i - \bar{y})(x_i - \bar{x})$$

$$\hat{w} = \frac{\sum_{i=1}^{N}(y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

1.4 Logistic regression

1.4.1 Probability basics

- $P(A)$: probability of event A.
- $P(A^c)$: probability of event A not happening.
- $P(A|B)$: probability of A given B (conditional probability).
- For any event A, $0 \leq P(A) \leq 1$.
- Let S be the sample space of all possible events, then $P(S) = 1$.
- For any event A, $P(A^c) = 1 - P(A) \iff P(A) = 1 - P(A^c)$.

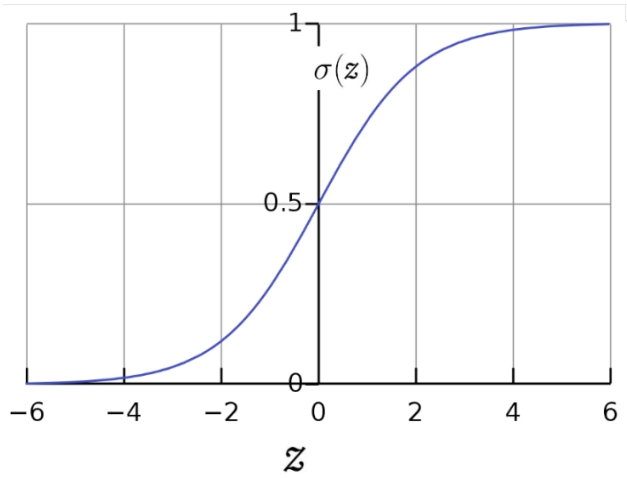
1.4.2 Discriminative classifier

Discriminative models learn boundaries between classes. For example, suppose we are classifying cat vs. dog images. The input x is the image (matrix of pixel values). The output y is the class (either cat or dog). The goal of classification is to find the correct class y given input x. To classify, we compute: $y = \underset{y \in C}{\text{argmax}} P(y|x)$.

1.4.3 Logistic regression

We introduce the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



The sigmoid function is called a logistic function. It squeezes any input value between 0 and 1. To classify the input $\mathbf{x}$:

- Compute the weighted sum, $z = \sum w_i x_i + b = \mathbf{w} \cdot \mathbf{x} + b$.
- Pass z through the sigmoid function, $\sigma(\mathbf{w} \cdot \mathbf{x} + b)$.
- Treat the result as a probability.

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w} \cdot \mathbf{x} + b) = \frac{1}{1 + \exp(-\mathbf{w} \cdot \mathbf{x} - b)}$$

A key property of the sigmoid function: $1 - \sigma(z) = \sigma(-z)$.

Derivation

$$\begin{aligned} P(y = 0|\mathbf{x}) &= 1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b) \\ &= 1 - \frac{1}{1 + \exp(-\mathbf{w} \cdot \mathbf{x} - b)} \\ &= \frac{\exp(-\mathbf{w} \cdot \mathbf{x} - b)}{1 + \exp(-\mathbf{w} \cdot \mathbf{x} - b)} \times \frac{\exp(\mathbf{w} \cdot \mathbf{x} + b)}{\exp(\mathbf{w} \cdot \mathbf{x} + b)} \\ &= \frac{1}{1 + \exp(\mathbf{w} \cdot \mathbf{x} + b)} \\ &= \sigma(-\mathbf{w} \cdot \mathbf{x} - b) \end{aligned}$$

Now, we have a way to compute both $P(y = 1|\mathbf{x})$ and $P(y = 0|\mathbf{x})$. We can now make decisions:

$$\text{decision}(\mathbf{x}) = \begin{cases} 1 & \text{if } P(y = 1|\mathbf{x}) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

The decision boundary of $P(y|x) = 0.5$ is defined where $\mathbf{w} \cdot \mathbf{x} + b = 0$. This is the default decision boundary in logistic regression. However, you can specify a different decision boundary to suit your task.

- Choose a higher value, like $P(y = 1|\mathbf{x}) > 0.9$, if you want the model to be confident before flagging an instance of a positive class.
- Choose a lower value, like $P(y = 1|\mathbf{x}) > 0.2$, if you don't want to risk missing instances of the positive cass (e.g. medical diagnosis).

1.4.4 Example

Predict if a fire weather warning should be announced.

- x_1: temperature in C°.
- x_2: wind speed in km/h.
- $y = 0$: no warning needed.
- $y = 1$: warning needed.

Suppose the weights are $\mathbf{w} = [0.1, 0.05]$ and the bias term is $b = -5$. For an input vector $\mathbf{x} = [25, 60]$,

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w} \cdot \mathbf{x} + b) = \sigma((0.1)(25) + (0.05)(60) - 5) = \sigma(0.5) = 0.622$$
$$P(y = 0|\mathbf{x}) = 1 - P(y = 1|\mathbf{x}) = 1 - 0.622 = 0.378$$

Since $P(y = 1|\mathbf{x}) = 0.622 > 0.5$, a warning should be given.

1.5 Cross-entropy loss

1.5.1 Log probabilities

Definition: Log probability

A log probability is a logarithm of a probability value, $\log P(X)$. This represents probabilities on a logarithmic scale $(-\infty, 0]$ instead of the usual $[0, 1]$ probability interval.

A useful property of log properties:

$$\log P(ABC) = \log(P(A) \times P(B) \times P(C)) = \log P(A) + \log P(B) + \log P(C)$$

1.5.2 Loss function

We use conditional **maximum likelihood estimation**:

- Find parameters that **maximize the probability of the true labels** in the training data given the observations.
- For binary classification, since there are 2 discrete outcomes, we can express the classifier's probability of the true label y as:

$$P(y|\mathbf{x}) = \hat{y}^y(1 - \hat{y})^{1-y}$$

- If $y = 1$, this simplifies to $\hat{y}$ which is $P(y = 1|\mathbf{x})$.
- If $y = 0$, this simplifies to $1 - \hat{y}$ which is $P(y = 0|\mathbf{x})$.

We can rewrite this in terms of log probabilities:

$$\begin{aligned}\log P(y|\mathbf{x}) &= \log\left[\hat{y}^y(1-\hat{y}^{1-y})\right] \\ &= y\log\hat{y} + (1-y)\log(1-\hat{y})\end{aligned}$$

We can then turn this into a loss function to be minimized by using the negative log likelihood. This is called the **cross-entropy loss** between the prediction and the true class.

$$\begin{aligned}L_{\text{CE}}(\hat{y}, y) &= -\log P(y|\mathbf{x}) \\ &= -\big[y\log\hat{y} + (1-y)\log(1-\hat{y})\big] \\ &= -\big[y\log[\sigma(\mathbf{w}\cdot\mathbf{x}+b)] + (1-y)\log[\sigma(-\mathbf{w}\cdot\mathbf{x}-b)]\big]\end{aligned}$$

By minimizing this loss, we maximize the probabilities assigned to the true labels by the logistic regression model.

1.5.3 Example

Predict if a fire weather warning should be announced.

- x_1: temperature in C°.
- x_2: wind speed in km/h.
- $y = 0$: no warning needed.
- $y = 1$: warning needed.

Suppose the weights are $\mathbf{w} = [0.1, 0.05]$ and the bias term is $b = -5$.

For an input vector $\mathbf{x} = [25, 60]$ and true label $y = 1$:

$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}\cdot\mathbf{x} + b) = 0.622$

$L_{\text{CE}}(\hat{y}, y) = -\log P(y = 1|\mathbf{x}) = -\log 0.622 = 0.206$

For an input vector $\mathbf{x} = [25, 60]$ and true label $y = 0$:

$P(y = 0|\mathbf{x}) = 1 - \sigma(\mathbf{w}\cdot\mathbf{x} + b) = 0.378$

$L_{\text{CE}}(\hat{y}, y) = -\log P(y = 0|\mathbf{x}) = -\log 0.378 = 0.423$

Since the loss of true label $y = 1$ is lower, we should give a warning.

Note

The loss is higher if the model is wrong. Lower probabilities assigned to the true class implies a higher loss.

1.5.4 Objective function

Given the cross-entropy loss, we want to minimize this loss on average across our training set:

$$\boxed{\frac{1}{N}\sum_{i=1}^{N} L_{\text{CE}}(\hat{y}_i, y_i)}$$

If we can adjust parameters $\mathbf{w}$ and b to lower the loss over all instances in the training set, we will have a good classifier. This is the essence of modern machine learning and deep learning.

1.5.5 Optimization

Let the loss function be parameterized by parameters $\theta = (\mathbf{w}, b)$. We represent $\hat{y}$ as $f(\mathbf{x}; \theta)$. We want to find parameters that minimize the loss averaged over all examples in our training set.

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^{N} L_{\text{CE}}(f(\mathbf{x}; \theta), y_i)$$

There is **no closed form solution** for finding minimizing weights for logistic regression. Instead, we will use an iterative method called gradient descent.

1.6 Gradient descent

Definition: Gradient descent

Gradient descent is an optimization algorithm that iteratively updates parameters by moving in the opposite direction of the gradient of the loss function, in order to minimize the loss.

1.6.1 Loss function

We want parameters $\mathbf{w}, b$ that minimize loss averaged over m examples in our training set. We denote the average loss as $J(\mathbf{w}, b)$ with $\hat{y}_i = f(\mathbf{x}_i; \mathbf{w}, b)$ written as a function of inputs and parameters.

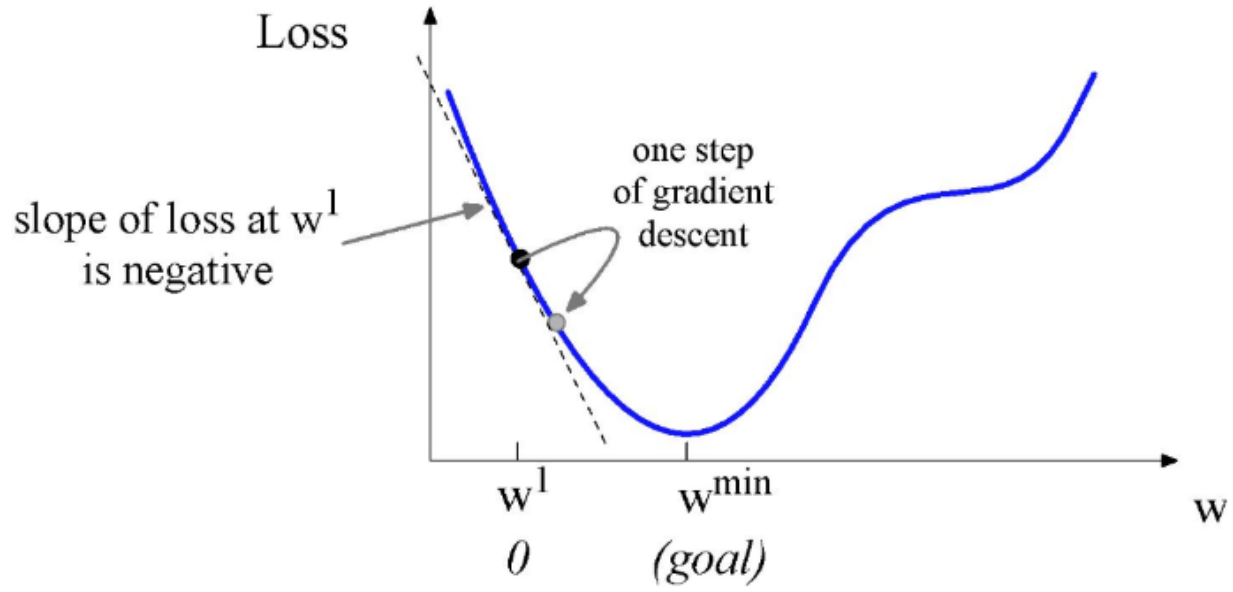
$$J(\mathbf{w}, b) = \frac{1}{m} \sum_{i=1}^{m} \left[y_i \log \sigma(\mathbf{w} \cdot \mathbf{x}_i + b) + (1 - y_i) \log[1 - \sigma(\mathbf{w} \cdot \mathbf{x}_i + b)] \right]$$

Now, we want to find:

$$\hat{\mathbf{w}}, \hat{b} = \underset{\mathbf{w}, b}{\operatorname{argmin}} J(\mathbf{w}, b)$$

1.6.2 Optimization

For logistic regression, the loss function is convex. Gradient descent from any point is guarenteed to find the minimum. Here is a visualization for a single scalar w: Given the current w, move in the reverse direction from the function slope.



The loss is a function of multiple parameters $w_1, \ldots, w_n, b$:

$$L_{\text{CE}} = -\big[y \log \sigma(\mathbf{w} \cdot \mathbf{x} + b) + (1 - y) \log[1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b)]\big]$$

The gradient of this loss with respect to the parameters θ will be a vector of partial derivatives.

$$\nabla_\theta L = \frac{\partial L}{\partial \theta} = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \vdots \\ \frac{\partial L}{\partial w_n} \\ \frac{\partial L}{\partial b} \end{bmatrix}$$

For models with n-dimensional weight vectors:

- For each dimension w_i, the gradient component i tells us the slope w.r.t that parameter.
- We express the slope as the partial derivative of L w.r.t w_i.
- The gradient is defined as the vector of these partial derivatives.

1.6.3 Learning rate

$$\mathbf{w}^{t+1} = \mathbf{w}^t - \eta \frac{d}{dw} L(f(\mathbf{x}; \mathbf{w}), y)$$

The gradient tells us:

- **Sign:** which way to move (left, right, up, down).
- **Magnitude:** how steep/big the update is.

To control the step size, the value of the gradient is weighted by a learning rate η. A higher value of η means $\mathbf{w}$ changes faster. This is considered a hyperparameter.

- If the learning rate is too high, it will take big steps and overshoot.
- If the learning rate is too low, it will take too long to converge.

1.6.4 Optimization (continued)

How do we find $\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \frac{1}{m} \sum_{i=1}^{m} L_{\text{CE}}(f(\mathbf{x}; \theta), y_i)$? We do the following iteratively:

- Compute the derivative of the loss function with respect to the parameters.
- Update the parameters to move in the opposite direction of the gradient.

The partial derivatives for logistic regression:

$$\boxed{\frac{\partial L}{\partial w_i} = [\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y] x_i} \qquad \boxed{\frac{\partial L}{\partial b} = \sigma(\mathbf{w} \cdot \mathbf{x} + b) - y}$$

1.6.5 Gradient descent formula

The final question for updating parameters θ based on the gradient is:

$$\theta^{t+1} = \theta^t - \eta \nabla_\theta L(f(\mathbf{x}; \theta), y)$$

Note

Gradient descent can be applied to any model with any number of parameters. Gradient descent is a general optimization method, not just for logistic regression!

1.7 Stochastic gradient descent

Definition: Stochastic gradient descent

Stochastic gradient descent (SGD) iteratively updates parameters. It approximates the true gradient by using a subset of data at each step.

At each step, a random subset of examples is sampled. Both the gradient is computed and the parameters are updated based on this subset.

$$\theta^{t+1} = \theta^t - \eta \frac{1}{B} \sum_{i=1}^{B} \nabla_\theta L(f(\mathbf{x}_i; \theta), y_i)$$

Batch / mini-batch: The subset of data used for SGD at a particular training step.
Batch size: The number of training examples in each batch. **This is a hyperparameter.**

1.7.1 Batching

In practice, it's computationally expensive to apply gradient descent to the entire training set simultaneously. This is due to the high computational/memory requirements for large datasets and the slow convergence. Instead, we employ the principle of batching.

Traverse the training set in batches, B examples at a time, updating parameters based on gradient w.r.t average loss of the batch.

$$J_B(\theta) = \frac{1}{B} \sum_{i=1}^{B} L(f(\mathbf{x}_i; \theta), y_i)$$

The gradient of the cross-entropy loss averaged over all B examples in a batch is the average of the individual sample gradients.

$$\nabla_\theta J_B(\theta) = \frac{1}{B}\sum_{i=1}^{B} \nabla_\theta L(f(\mathbf{x}_i;\theta), y_i)$$

1.7.2 Example

Assume initial parameters θ^0 are all zero: $w_1 = w_2 = b = 0$. let's perform a SGD update based on one example in the training set. Let $\mathbf{x} = [3, 2]$ with label $y = 1$. Then,

$$\begin{aligned}
\nabla_\theta = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \frac{\partial L}{\partial b} \end{bmatrix} &= \begin{bmatrix} [\sigma(\mathbf{w}\cdot\mathbf{x}+b) - y]x_1 \\ [\sigma(\mathbf{w}\cdot\mathbf{x}+b) - y]x_2 \\ [\sigma(\mathbf{w}\cdot\mathbf{x}+b) - y] \end{bmatrix} \\
&= \begin{bmatrix} [\sigma(0) - 1](3) \\ [\sigma(0) - 1](2) \\ [\sigma(0) - 1] \end{bmatrix} \\
&= \begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix}
\end{aligned}$$

Now that we have the gradient vector, we compute the new parameter values with gradient descent. Set the learning rate $\eta = 0.1$.

$$\begin{aligned}
\theta^1 &= \theta^0 - \eta\nabla_\theta L \\
&= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - (0.1)\begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix} \\
&= \begin{bmatrix} -0.15 \\ -0.10 \\ -0.05 \end{bmatrix}
\end{aligned}$$

1.7.3 Vectorization

Construct matrices and vectors that allow us to process a batch as a matrix/vector computation. Let the rows of matrix X represent input features of individual trainign samples with B examples in the batch and n input features:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{B1} & x_{B2} & \cdots & x_{Bn} \end{bmatrix}_{B\times n}$$

The vector $\mathbf{w}$ is the weight parameters. Let $\mathbf{b}$ be a B-dimensional vector repeating bias parameter b.

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}_{n\times 1} \qquad \mathbf{b} = \begin{bmatrix} b \\ b \\ \vdots \\ b \end{bmatrix}_{B\times 1}$$

Then we can compute the weighted sums $\mathbf{w} \cdot \mathbf{x}_i + b$ simultaneously for all B examples with matrix multiplication.

$$X\mathbf{w} + \mathbf{b} = \begin{bmatrix} w_1 x_{11} + w_2 x_{12} + \cdots + w_n x_{1n} + b \\ w_1 x_{21} + w_2 x_{22} + \cdots + w_n x_{2n} + b \\ \vdots \\ w_1 x_{B1} + w_2 x_{B2} + \cdots + w_n x_{Bn} + b \end{bmatrix}_{B \times 1}$$

If we aply the sigmoid function to each element of this vector:

$$\hat{\mathbf{y}} = \sigma(X\mathbf{w} + \mathbf{b}) = \begin{bmatrix} \sigma(w_1 x_{11} + w_2 x_{12} + \cdots + w_n x_{1n} + b) \\ \sigma(w_1 x_{21} + w_2 x_{22} + \cdots + w_n x_{2n} + b) \\ \vdots \\ \sigma(w_1 x_{B1} + w_2 x_{B2} + \cdots + w_n x_{Bn} + b) \end{bmatrix}_{B \times 1} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_B \end{bmatrix}$$

1.8 Multinomial logistic regression

1.8.1 One-vs-rest classification

How can we generalize logistic regression to multi-class regression? **One-vs-rest classification**.

- Train K binary logistic regression models.
- Each classifier is trained to classify one class against all other classes.
- To make a prediction for a data point:
  - All K classifiers are used.
  - Each gives us a probability $P(y = 1|\mathbf{x})$ of its target class.
  - Take the class with the highest probability as the prediction.

For each class $c \in \{1, \ldots, K\}$, we have a logistic regression model with its own weight vector $\mathbf{w}_c$ and bias term b_c. It is trained to maximize $P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}_c \cdot \mathbf{x} + b_c)$ for class c.

Note

Because the sigmoid function is monotonically increasing, we don't need the probabilities. The weighted sums are enough!

Higher $\mathbf{w}_c \cdot \mathbf{x} + b_c \Rightarrow$ Higher $\sigma(\mathbf{w}_c \cdot \mathbf{x} + b_c)$.

To make a final decision, we compute all linear scores:

$$s_c(\mathbf{x}) = \mathbf{w}_c \cdot \mathbf{x} + b_c \quad \text{for each class } c \in \{1, \ldots, K\}$$

and then choose the predicted class as the one with the largest score:

$$\hat{y} = \underset{c \in \{1,\ldots,K\}}{\operatorname{argmax}} \; s_c(\mathbf{x})$$

where $\hat{y}$ is the predicted class.

Disadvantages

We need to train K different model for one task. In return, we get K unrelated probabilities that don't sum to 1.

1.8.2 Multinomial logistic regression

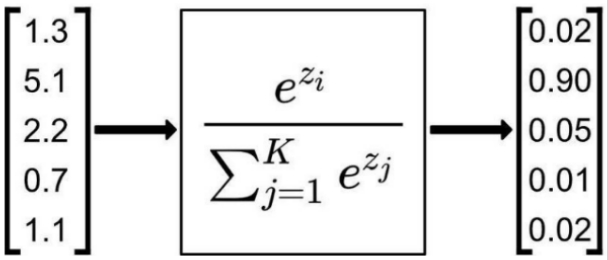
For multi-class classification, we have $y \in \{1, \ldots, K\}$:

$$\begin{bmatrix} P(y=1|\mathbf{x}) \\ P(y=2|\mathbf{x}) \\ \vdots \\ P(y=K|\mathbf{x}) \end{bmatrix}$$

So instead of a single probability for the positive class, we need to output probabilities for each class: $P(y=1|\mathbf{x}), \ldots, P(y=K|\mathbf{x})$ with

$$\boxed{\sum_{k=1}^{K} P(y=k|\mathbf{x}) = 1}$$

Instead of the sigmoid function, we can use the **softmax function** to squeeze K arbitrary values into probabilities summing to 1.



For input $\mathbf{x}$, the score (also called **logit**) for class c is $z_c = \mathbf{w}_c \cdot \mathbf{x} + b_c$. But probabilities need to be positive, so take the exponential e^{z_c}. But probabilities also need to sum to 1, so normalize.

The softmax function:

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{k=1}^{K} \exp(z_k)} \quad \text{for } 1 \le i \le K$$

This gives the softmax regression model:

$$f(\mathbf{x}; W, b) = \begin{bmatrix} P(y=1|\mathbf{x}; W, b) \\ P(y=2|\mathbf{x}; W, b) \\ \vdots \\ P(y=K|\mathbf{x}; W, b) \end{bmatrix} = \frac{1}{\sum_{k=1}^{K} \exp(\mathbf{w}_k^\top \cdot \mathbf{x} + b_k)} \begin{bmatrix} \exp(\mathbf{w}_1^\top \cdot \mathbf{x} + b_1) \\ \exp(\mathbf{w}_2^\top \cdot \mathbf{x} + b_2) \\ \vdots \\ \exp(\mathbf{w}_K^\top \cdot \mathbf{x} + b_K) \end{bmatrix} = \text{softmax}(W\mathbf{x}+\mathbf{b})$$

The model parameters are W and $\mathbf{b}$. W is a matrix with each row being the weight vector for a class. $\mathbf{b}$ is a vector of bias terms for each class.

$$W = \begin{bmatrix} \mathbf{w}_1^\top \\ \mathbf{w}_2^\top \\ \vdots \\ \mathbf{w}_K^\top \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_K \end{bmatrix}$$

Together, $\mathbf{w}_k$ and b_k define the linear decision boundary for class k.

The cross-entropy loss is also generalized to multiple classes. Now, $\hat{y}$ is a vector of K probabilities. Let y be a vector of targets, setting $y_c = 1$ for the true class and $y_c = 0$ for all other classes.

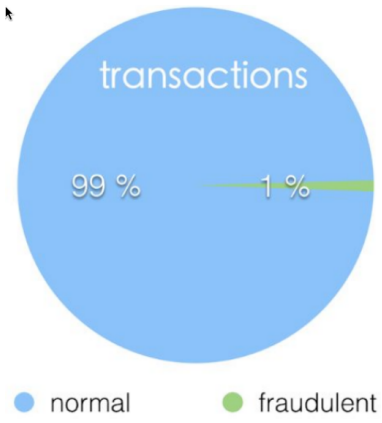
$$L_{\text{CE}} = -\sum_{k=1}^{K} y_k \log \hat{y}_k = -\sum_{k=1}^{K} y_k \log P(y = k|\mathbf{x})$$

This loss corresponds to what is called the negative log-likelihood. Without the minus sign, this is once again what we want to maximize with maximum likelihood estimation. For multinomial logistic regression, the cross-entropy loss is a function of all parameters $W, \mathbf{b}$.

$$\begin{aligned} L_{\text{CE}} &= -\sum_{k=1}^{K} y_k \log \hat{y}_k \\ &= -\log \hat{y}_c \quad \text{(where } c \text{ is the correct class)} \\ &= -\log \hat{P}(y_c = 1|\mathbf{x}) \\ &= -\log \frac{\exp(\mathbf{w}_c \cdot \mathbf{x} + b_c)}{\sum_{k=1}^{K} \exp(\mathbf{w}_k \cdot \mathbf{x} + b_k)} \end{aligned}$$

1.8.3 Imbalanced classes

For binary classification, the problem is framed as mapping inputs x to positive or negative labels. In a **balanced dataset**, both labels appear in **roughly equal numbers**. In an **imbalanced dataset**, **one label occurs much more often** than the other. For example, the training set of transactions labelled fraudulent vs. legitimate can have much higher proportion of legitimate transactions.



The problem is, does your dataset contain enough minority class examples to train a model properly?

- For mild imbalance (20%-40% minority) and sometimes for moderate imbalance (1%-20%), imbalance isn't a problem.
- For extreme imbalance (<1% minority), the model trains almost exclusively on one label and can't learn enough about the other label.

1.8.4 Handling imbalanced datasets

Downsampling majority class:
Reduce the number of majority examples by randomly selecting a subset of all majority examples to train on. For example, 1:10 → downsample by 1/5 times → 1:2. You can perform the sampling before training starts or at the start of every epoch.

Upsampling minority class:
Keep all majority class examples but duplicate minority class to match majority class size. For example, 1:10 → upsample by 10 times → 1:1.

Warning

This can lead to an **increase in the risk of overfitting** since the same minority examples are repeated multiple times!

Upsampling/downsampling is something you have to...

1. **Apply carefully:** it can improve performance, but it is not guaranteed to. It has trade-offs e.g. overfitting with upsampling and data loss with downsampling.
2. **Experiment with sampling ratios:** try different up/downsampling factors, e.g. 1:2, 1:3, 1:10, rather than assuming a perfectly balanaced dataset is alwasy best.
3. **Validate on a held-out validation set:** check that performance gains generalize beyond the resampled training data.

1.9 Text classification

Given a document d and a fixed set of classes $C = \{c_1, c_2, \ldots, c_K\}$, can we accurately predict the true class $c \in C$?

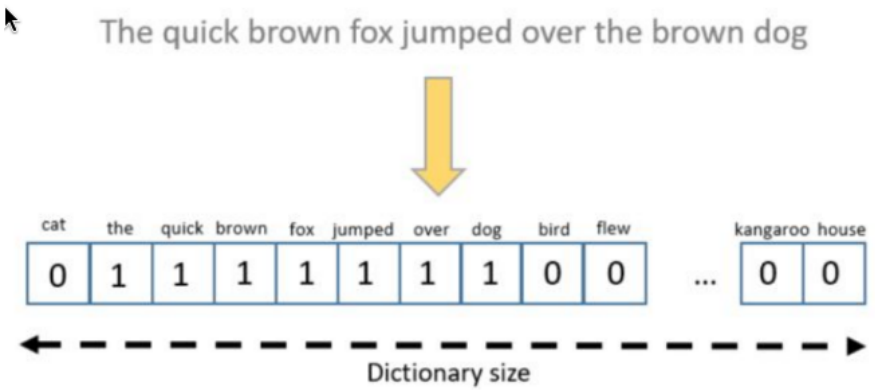
Rule-based methods:
These are hard-coded rules based on combinations of words or other related features. The accuracy can be high if the rules are carefully refined by the expert(s), but building and maintaining these rules is expensive.

Machine learning methods:
For machines, given data D where inputs are documents and the outputs are classes, it can effectively learn a classifier $h : X \to Y$.

1.9.1 Bag-of-words

In a bag-of-words approach, we represent documents with word counts ignoring order.

1. Tokenize text into words and assign an integer ID to each unique word in the training dataset, called the vocabulary.
2. Count the word occurences in each document.
3. Each document becomes a vector of indexed counts.



Word counts can be informative features for classification tasks. In logistic regression, each feature (word) will have a weight parameter w_i, which encodes its influence on predictions.

vocab = [a, boring, great, is, love, terrible, the, ...]
$\boldsymbol{w}$ = [0.0, -2.2, +2.5, 0.0, +2.0, -2.5, 0.0, ...], $b = 0$

Positive review: "the movie is great and I love it"

- Feature vector = [0, 0, 1, 1, 1, 0, 1, ...]
- $\boldsymbol{w} \cdot \boldsymbol{x} + b = 1\times(+2.5) + 1\times(+2.0) = +4.5$
- $\sigma(4.5) \approx 0.99 \Rightarrow$ predict positive

Negative review: "the movie is boring and terrible"

- Feature vector = [0, 1, 0, 1, 0, 1, 1, ...]
- $\boldsymbol{w} \cdot \boldsymbol{x} + b = 1\times(-2.2) + 1\times(-2.5) = -4.7$
- $\sigma(-4.7) \approx 0.009 \Rightarrow$ predict negative

Most words never appear in a given document.

- Vocabulary can be very large (e.g. 100k words).
- Each document uses only a small fraction (100-1000 words), so most vectors are mostly zero (>99%).
- It is common to use sparse data structures for storage and efficiency.

Design choices affect what ends up in your vocabulary.

- **Text preprocessing/cleaning:** lowercase words, remove puncuation or number if not useful.
- **Tokenization:** the simple, straightforward approach is to split on whitespace characters. There are more sophisicated tokenizers available.
- **Vocabulary size control:** *(based on training data frequencies!)*
  - Keep only top N frequent words (tune N).
  - Don't include rare words (occurs less than some frequency threshold).

Tradeoff

A bigger vocabulary could produce better features, but might result in slower training!

1.9.2 Binary presence

Another option is to simplify bag-of-words to presence/absence of words.

- Feature = 1 if word appears, 0 otherwise.
- Ignores frequency of word within document.
- Useful when word presence is more informative than counts.

1.9.3 TF-IDF

The problem is that raw words counts are misleading because common words like "the" or "is" appear everywhere but add zero meaning. Truly informative words (e.g "remarkable", "special", "uninspiring") are rare and give a document its unique identity. We can use TF-IDF (term frequency - inverse document frequency) to scale down frequent, generic words and scale up rare, highly descriptive words to surface the most meaningful features.

TF (term frequency): number of times term t appears in document d divided by total number of terms in document d.

IDF (inverse document frequency):

- $\text{idf}(t) = \log(N/\text{df}(t))$, where
- N is the number of documents in the training set, $\text{df}(t)$ id the number of documents containing term t.
- $\text{idf}(t)$ is low if term t is common and high when term t is rare.

TF-IDF: tf $\times$ idf

Extra

Usually the TF-IDF vector is L2-normalized. Reminder on L2-normalization:

$$x_{ij}^{\text{new}} = \frac{x_{ij}}{\sqrt{\sum_{i=1} x_{ij}^2}}$$

1.9.4 Example 1

Suppose we have 100 documents of 200 words each. The term "the" appears 20 times in document d and in 90 documents overall.

- tf("the", d) = 20/200 = 0.1
- idf("the") = log(100/90) = 0.046
- tf-idf("the", d) = 0.1 × 0.046 = 0.0046

1.9.5 Example 2

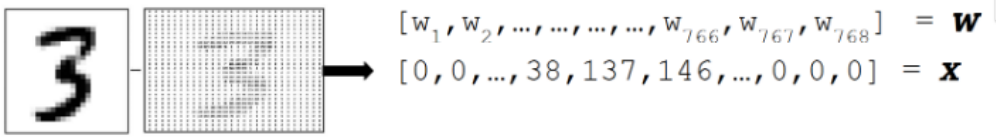
Suppose we have 100 documents of 200 words each. The term "happy" appears 5 times in document d and in 5 documents overall.

- tf("happy", d) = 5/200 = 0.025
- idf("happy") = log(100/5) ≈ 3
- tf-idf("happy", d) = 0.025 × 3 = 0.075

1.10 Regularization

1.10.1 Interpreting logistic regression

After training, learned weights capture how each feature contributes to model predictions, giving us insight into what the model has learned.

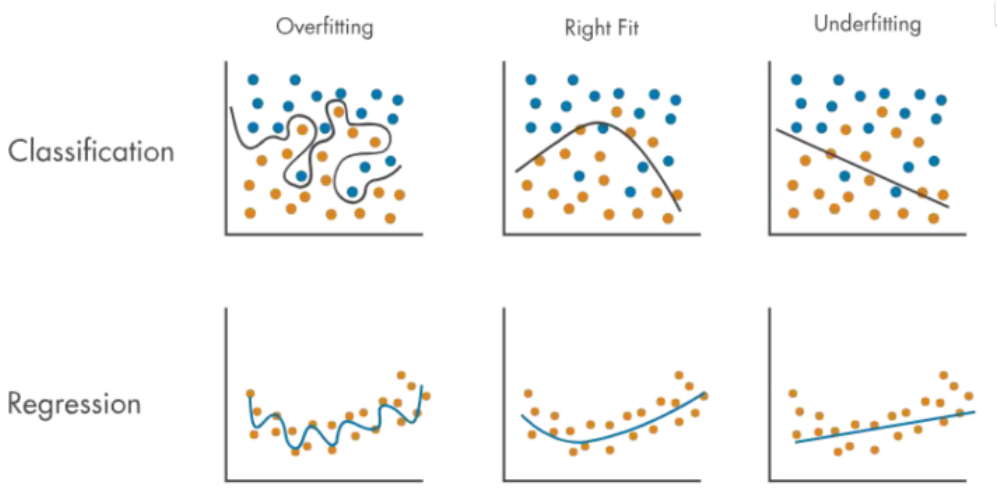


The magnitude of a weight $|w_i|$ reflects how strongly the associated feature influences the final predictions. Larger values have a stronger effect. The sign of w_i shows the direction of influence. Positive weights push the prediction towards class 1, while negative weights push the prediction towards class 0.

1.10.2 Overfitting

Definition: Overfitting

Overfitting is when your model is fitting the details of the training data so closely that the model doesn't generalize well to the test set. This happens when the model is over-parameterized. In other words, it has more parameters than required to model the data. The model is "too powerful", so it memorizes the training data. It predicts training data perfectly, but cannot generalize to new data.



How can overfitting happen for logistic regression?

- The weight w_i is learned such that $w_i x_i$ contributes to improved accuracy.
- If a feature x_i occurs only in one class in the training data, the model could learn very high weight w_i and perfectly predict y.
- The model might be over-relying on the presence of this feature, due to a peculiar property of the training data.
- The model will fail to generalize to a test set without this feature.

Tracking performance

High training accuracy, but low validation accuracy OR low training loss, but high validation loss.

- That's why it's important to track validation perfroamcne during training!
- If you only track training performance, this tells you nothing about how your model will generalize.

Potential causes of overfitting:

- **Wrong model for task:** model complexity exceeds available data.
- **Training data too small:** model will memorize rather than generalize.
- **Low-quality training data:** too much noise, which the model learns as if it is signal noise.
- **Too many training epochs:** eventually model fits training set too closely.

1.10.3 Early stopping

Definition: Early stopping

Stop training when the performance on held-out dataset stops improving to avoid overfitting.

Validation performance usually improve before degrading with overfitting. After this, improving training data fit comes at expense of generalization. Continually monitor performance on validation dataset and stop training where validation accuracy peaks.

1.10.4 Regularization

Add a regularization term $R(\theta)$ to the objective function.

$$\hat{\theta} = \underset{\theta}{\text{argmax}}\Big[\sum_{i=1}^{B} \log P(y_i|\mathbf{x}_i;\theta) - \alpha R(\theta)\Big]$$

Choose $R(\theta)$ to penelize large parameters θ. Loss favors predicting data less well with small θ than predicting data well with big θ. More formally, we force optimization to balance maximizing the log-likelihood while minimizing weight magnitudes.

1.10.5 L2 Regularization

The regularization term is the sum of the squares of the weights. The L2 norm $||\theta||_2$ is the Euclidean distance of θ to the origin.

$$R(\theta) = ||\theta||_2^2 = \sum_{i=1}^{n} \theta_i^2$$

The L2 regularized log-likelihood:

$$\hat{\theta} = \underset{\theta}{\text{argmax}}\Big[\sum_{i=1}^{B} \log P(y_i|\mathbf{x}_i;\theta) - \alpha \sum_{i=1}^{n} \theta_n^2\Big]$$

This is equivalent to minimizing the cross-entropy loss plus the L2 norm.

$$\hat{\theta} = \underset{\theta}{\text{argmax}}\Big[\frac{1}{B}\sum_{i=1}^{B} L_{\text{CE}}(\hat{y}_i, y_i) + \alpha \sum_{i=1}^{n} \theta_n^2\Big]$$

1.10.6 L1 Regularization

The regularization term is the sum of the absolute values of the weights. The L1 norm $||\theta||$ is the Euclidean distance of θ to the origin.

$$R(\theta) = ||\theta||_1 = \sum_{i=1}^{n} |\theta_i|$$

The L1 regularized log-likelihood:

$$\hat{\theta} = \underset{\theta}{\text{argmax}}\Big[\sum_{i=1}^{B} \log P(y_i|\mathbf{x}_i;\theta) - \alpha \sum_{i=1}^{n} |\theta_n|\Big]$$

This is equivalent to minimizing the cross-entropy loss plus the L1 norm.

$$\hat{\theta} = \underset{\theta}{\text{argmax}}\Big[\frac{1}{B}\sum_{i=1}^{B} L_{\text{CE}}(\hat{y}_i, y_i) + \alpha \sum_{i=1}^{n} |\theta_n|\Big]$$

1.10.7 Regularization (continued)

L2 regularization produces weight vectors with many small weights:

$$[0.05, -0.01, 0.01, -0.005, 0.1, \dots]$$

L1 regularization tends to produce sparse solutions with some larger weights but many more weights set to zero.

$$[1.1, 0.0, -2.5, 0.0, 0.0, \dots]$$

$$\hat{\theta} = \underset{\theta}{\text{argmax}} \Big[\sum_{i=1}^{B} \log P(y_i|\mathbf{x}_i; \theta) - \alpha R(\theta) \Big]$$

- α is a hyperparameter that controls the strength of regularization.
- During training, the model is baised towards smaller parameter values, which prevent over-reliance on certain features.
- These features could be unique to the training data, so not relying on them can reduce overfitting.

1.11 Naive Bayes

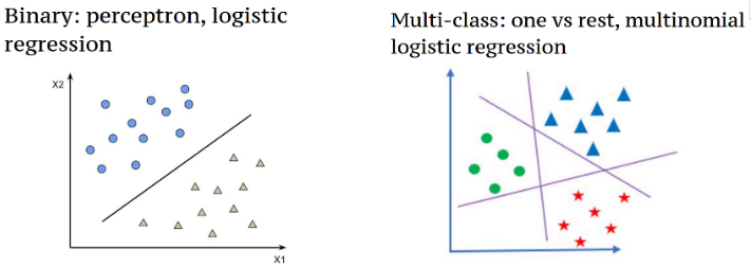
1.11.1 Generative vs. discriminative classifiers

Generative models learn the distribution of individual classes $P(x|y)$. Model how data is generated. Given a new example to classify, based on generation assumption, which category is most likely to generate this data?

Discriminative models learn the boundaries between classes $P(y|x)$. Model decision boundary between classes rather than underlying distribution of each class. Learns defining differences between classes and classifies examples based on these differences.

Discriminative classifier: estimate the posterior $P(y|x)$ directly. This is what logistic regression computes.

$$y = \underset{y \in C}{\text{argmax}} \overbrace{P(y|x)}^{\text{posterior}}$$



Generative classifier: estimate likelihood $P(x|y)$ and apply Bayes' rule.

$$y = \underset{y \in C}{\text{argmax}} \overbrace{P(x|y)}^{\text{likelihood}} \overbrace{P(y)}^{\text{prior}}$$

1.11.2 Independence

Definition: Independence of events

Two events X and Y are independent when knowing one tells you nothing about the other. In other words, learning Y does not change the probability of X.

That is,

$$P(X|Y) = P(X) \iff P(Y|X) = P(Y)$$

If two events are independent, then te following holds:

$$P(X, Y) = P(X)P(Y)$$

1.11.3 Bayes' rule

Two ways to factor a join distribution over two variables:

$$P(x, y) = \frac{P(y)P(x|y)}{P(x)}$$

Dividing, we get Bayes' rule:

$$\boxed{P(y|x) = \frac{P(y)P(x|y)}{P(x)}}$$

where:

- $P(y|x)$ is the **posterior probability**.
- $P(y)$ is the **prior**.
- $P(x)$ is the **evidence**.
- $P(x|y)$ is the **likelihood**.

1.11.4 Naive Bayes classifier

Naive Bayes is a simple classification method based on Bayes' rule. For a document d, Bayes' rule can be used to calculate $P(c|d)$ for every output class $C = \{c_1, c_2, \ldots, c_K\}$.

$$P(c|d) = \frac{P(c)P(d|c)}{P(d)}$$

To obtain a prediction, find the class for which $P(c|d)$ is highest:

$$c_{\text{MAP}} = \underset{c \in C}{\text{argmax}}\, P(c|d)$$

MAP stand for "maximum a posteriori", or in other words most likely class.

Derivation

$$
\begin{aligned}
c_{\text{MAP}} &= \underset{c \in C}{\text{argmax}}\, P(c|d) \\
&= \underset{c \in C}{\text{argmax}}\, \frac{P(c)P(d|c)}{P(d)} \quad \text{(Bayes' rule)} \\
&= \underset{c \in C}{\text{argmax}}\, P(d|c)P(c) \quad \text{(Drop denominator, argmax unaffected)} \\
&= \underset{c \in C}{\text{argmax}}\, P(x_1, x_2, \ldots, x_n|c)P(c) \quad \text{(Represent document as features)}
\end{aligned}
$$

How can we estimate these probabilities? The simple solution is to just count how often each class occurs and then normalize.

Bag-of-words assumption: Assume word positions doesn't matter.

Conditional independence: Assume the feature probabilities $P(x_i|c_i)$ are independent given the class c:

$$P(x_1, x_2, \ldots, x_n|c) = P(x_1|c) \times P(x_2|c) \times \cdots \times P(x_n|c)$$

We can estimate these probabilities:

$$\hat{P}(w_i|c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

In other words, the relative frequency of w_i within class c_j. We can estimate $P(c)$ in the following way:

$$\hat{P}(c_j) = \frac{N_{cj}}{N_{\text{total}}}$$

where N_{cj} is the number of training examples in class c_j and N_{total} is the total number of training examples. Given our assumption, we can rewrite the prediction for Naive Bayes as:

$$
\begin{aligned}
c_{\text{NB}} &= \underset{c \in C}{\text{argmax}}\, P(c_j) \prod_{x \in X} P(x|c) \\
&= \underset{c \in C}{\text{argmax}}\, P(c_j) \prod_{i \in \text{positions}} P(x_i|c_j)
\end{aligned}
$$

where $i \in$ positions goes through all words in the document. While this is great, it comes with an implementation problem. Multiplying lots of probabilities can results in floating-point underflow. To combat this, simply use log probabilities!

$$
\begin{aligned}
c_{\text{NB}} &= \underset{c \in C}{\text{argmax}}\, P(c_j) \prod_{i \in \text{positions}} P(x_i|c_j) \\
&= \underset{c \in C}{\text{argmax}} \Big[\log P(c_j) + \sum_{i \in \text{positions}} \log P(x_i|c_j) \Big]
\end{aligned}
$$

1.11.5 Naive Bayes classifier: learning

The problem with maximum likelihood is the zero counts. What is we have seen no training documents with the word "fantastic" and labelled as "positive"? If any of the probability terms are 0,

the whole probability score will be 0.

Solution: Laplace (add-1) smoothing
Add 1 to the count of each (w, c):

$$\hat{P}(w_i|c) = \frac{\text{count}(w_i, c) + 1}{\sum_{w \in V} \text{count}(w, c) + |V|}$$

where $|V|$ is the size of the vocabulary. Unknown words get a tiny probability. Frequent words still dominate.

Solution: Additive smoothing
Add α to the count of each (w, c):

$$\hat{P}(w_i|c) = \frac{\text{count}(w_i, c) + \alpha}{\sum_{w \in V} \text{count}(w, c) + \alpha|V|}$$

Here, α is the smoothing parameter. It controls how much probability mass you give to unseen events. If $\alpha = 1$, that's classic Laplace smoothing.

Training algorithm:

1. From training corpus, extract the vocabulary.
2. Calculate $P(c_j)$ terms.
3. Calculate $P(w_k|c_j)$ terms.

Extra

What about unknown words that appear in the test data but not in the training data? It doesn't necessarily help. Knowing which class has more unkown words is not generally helpful.

1.11.6 Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable ~~with~~ no fun

1. Create vocabulary: 20 unique words in training data.

2. Calculate the priors from training: $\hat{P}(c_j) = N_{cj}/N_{\text{total}}$

$$P(-) = 3/5 \qquad P(+) = 2/5$$

3. Calculate likelihoods from training (Laplace smoothing). Here we have 14 tokens in negative docs and 9 in positive docs.

$$P(\text{"predictable"}|-) = \frac{1+1}{14+20} \qquad P(\text{"predictable"}|+) = \frac{0+1}{9+20}$$
$$P(\text{"no"}|-) = \frac{1+1}{14+20} \qquad P(\text{"no"}|+) = \frac{0+1}{9+20}$$
$$P(\text{"fun"}|-) = \frac{0+1}{14+20} \qquad P(\text{"fun"}|+) = \frac{1+1}{9+20}$$

4. Scoring the test example:

$$P(-)P(\text{"predicatble"}|-)P(\text{"no"}|-)P(\text{"fun"}|-) = 0.6 \times \frac{4}{32^2} = 6.1 \times 10^{-5}$$
$$P(+)P(\text{"predicatble"}|+)P(\text{"no"}|+)P(\text{"fun"}|+) = 0.4 \times \frac{2}{29^2} = 3.2 \times 10^{-5}$$

1.11.7 Binary Naive Bayes

For tasks like sentiment analysis, word occurence can be more important than word frequency. For example, the occurence of the word "fantastic" tells us a lot. The fact that it occurs 5 times may not tell us much more. Binary Naive Bayes is a variant of Naive Bayes:

- Clip our word counts at 1 for each document.
- Can implement by removing duplicates in each document and executing Naive Bayes algorithm as before.

Four original documents:

- − it was pathetic the worst part was the boxing scenes
- − no plot twists or great scenes
- + and satire and great plot twists
- + great scenes great film

After per-document binarization:

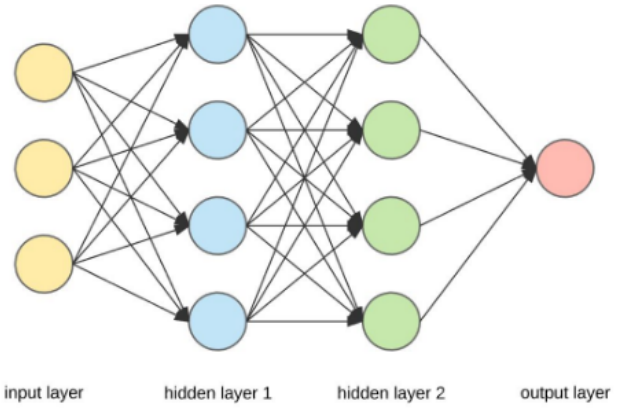
- − it was pathetic the worst part boxing scenes
- − no plot twists or great scenes
- + and satire great plot twists
- + great scenes film

	NB Counts		Binary Counts	
	+	−	+	−
and	2	0	1	0
boxing	0	1	0	1
film	1	0	1	0
great	3	1	2	1
it	0	1	0	1
no	0	1	0	1
or	0	1	0	1
part	0	1	0	1
pathetic	0	1	0	1
plot	1	1	1	1
satire	1	0	1	0
scenes	1	2	1	2
the	0	2	0	1
twists	1	1	1	1
was	0	2	0	1
worst	0	1	0	1

2 Artificial Neural Networks

2.1 Neural networks

Artificial neural networks (ANNS) consist of artificial neurons, much like brain cells, and are connected to each other via synapses. The artificial neurons pass signals (real numbers) between each other. Given some input, we would like to make a decision. Typically, we use non-linear activation functions for each neuron.



2.1.1 Neural network architecture

A neural network consists of:

- An input layer
- Hidden layer(s)
- An output layer

Each unit layer computes a value based on the linear combination of values of units to that point into it. What happens if you add more hidden layers? Increase the *depth* of the neural network.

2.2 Feedforward propagation formulation

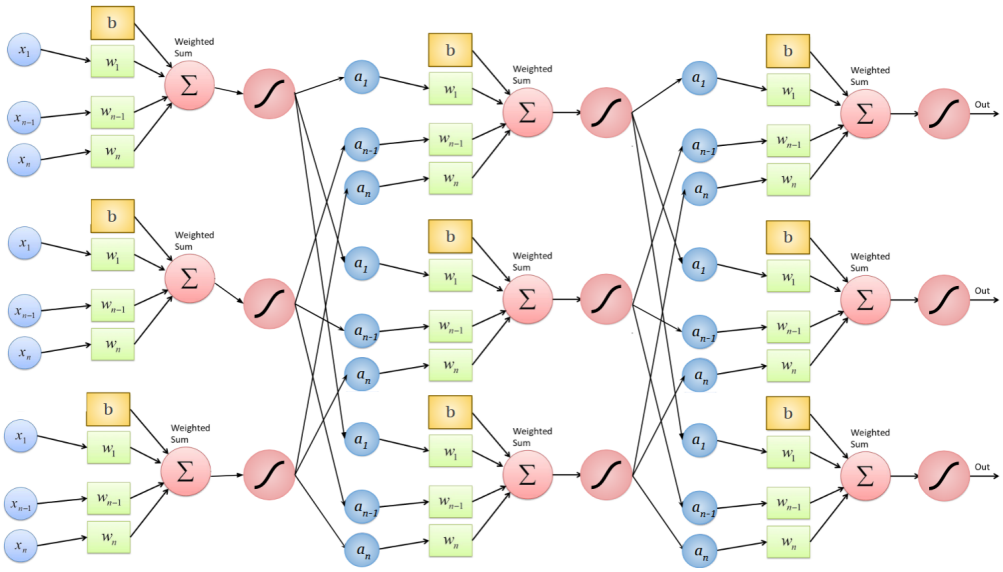
2.2.1 Activation functions

Definition: Activation function

The mathematical equation inside a neuron that determines the neuron's output.

What is the role of the activation function? It is used to introduce non-linearity into the neural network. How do they even work? Every neuron has two stages:

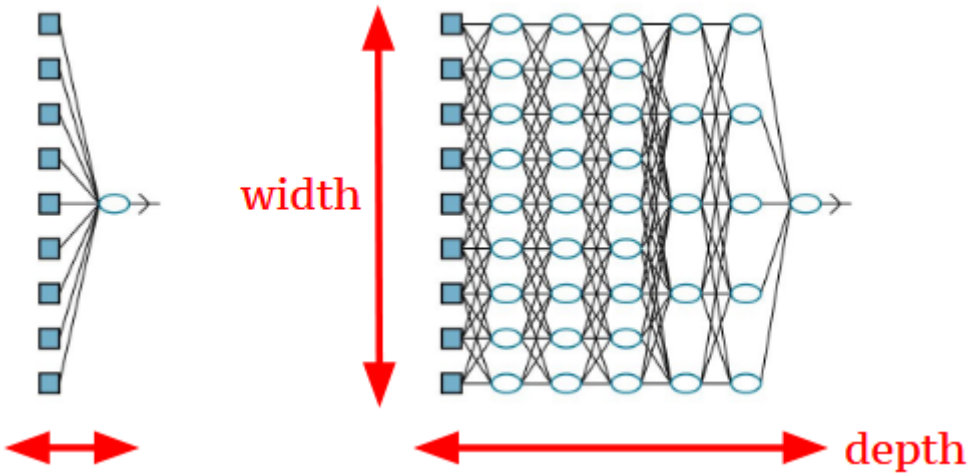
1. **Weighted sum:** The neuron receives inputs from the previous layer. Each input is multiplied bt a weight. These are all summed up, and a bias term is added. This results in a single number z.
2. **Activation:** This number z is then passed through the activation function, say $f(z)$, to produce the neuron's final output y. This output is then sent as an input to the neurons in the next layer.



Note

Every layer in a neural network has its own activation function.

2.2.2 Network structure

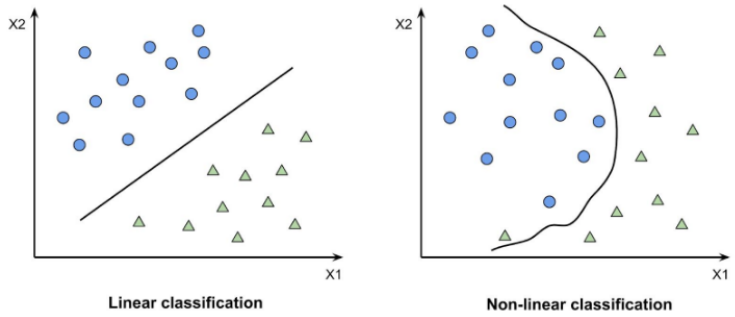


A **feedforward network** has connections going in one direction. Logistic regression can be viewed as ashallow (1-layer) neural network. Let each connection (line) represent a weight multiplication with the preceding unit output, and all incoming lines represent a weighted sum. Each unit computes: $a_j = g_j(\mathbf{w}_j^\top \mathbf{x} + b_j)$

- A weighted sum of its weights with the output of the previous layer.
- Passes that through an activation function g.

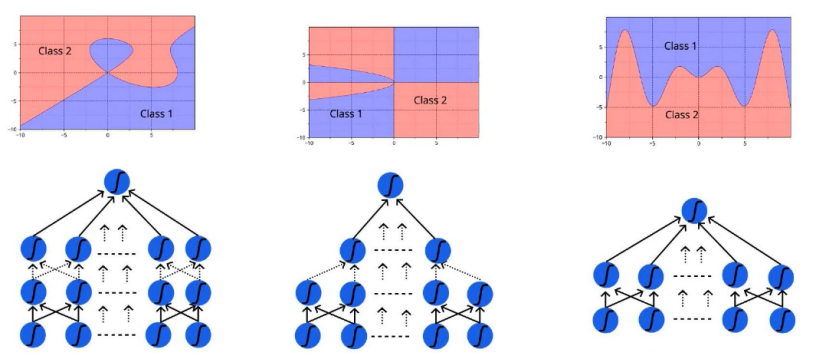
2.2.3 Non-linearity

Each unit j has a **nonlinear** activation function g_j through which the weighted sum is passed. Without nonlinearity, any composition of units would still represent a linear function. The nonlinearity allows sufficiently large networks of units to approximate arbitrary functions. Unlike linear regression, perceptrons, and logistic regression neural networks can model non-linear functions for regression or classification.



Theorem: Universal approximation theorem

A network with just 2 layers, the first non-linear and the second linear, can approximate any continuous function to arbitrary degree of accuracy, including non-linear function.



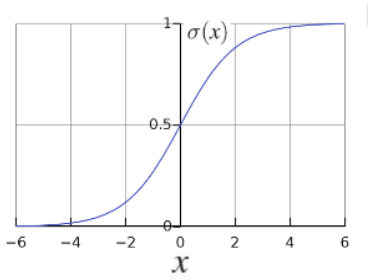
2.3 Activation functions

All of the following are monotonically non-decreasing (positive gradients). The choice of activation function is a modelling decision. It depends on factors like the task, the type of neural network, etc.

2.3.1 Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

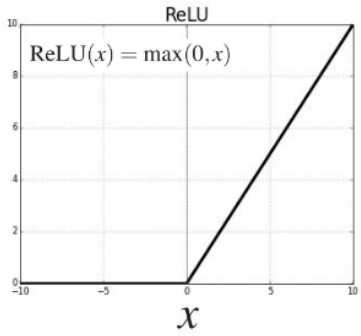
As seen with logistic regression.



2.3.2 ReLU

$$\text{ReLU}(x) = \max(0, x)$$

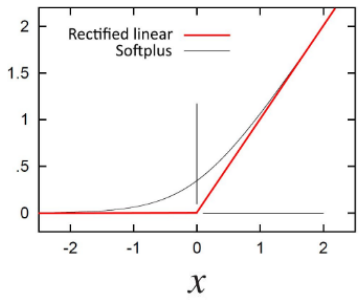
Non-linear function (gradient is not constant) but acts linear for part of the domain (>0). This is a very commonly used activation function.



2.3.3 Softplus

$$\text{softplus}(x) = \log(1 + e^{x})$$

The smooth version of ReLU function.

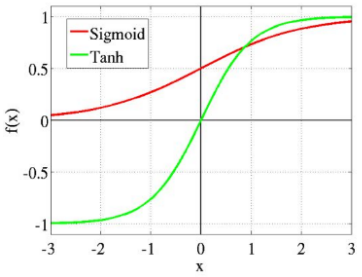


2.3.4 Tanh

$$\tanh(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$

Range of tanh is $[-1, 1]$. It is the scaled and shifted version of the sigmoid,

$$\tanh(x) = 2\sigma(2x) - 1$$

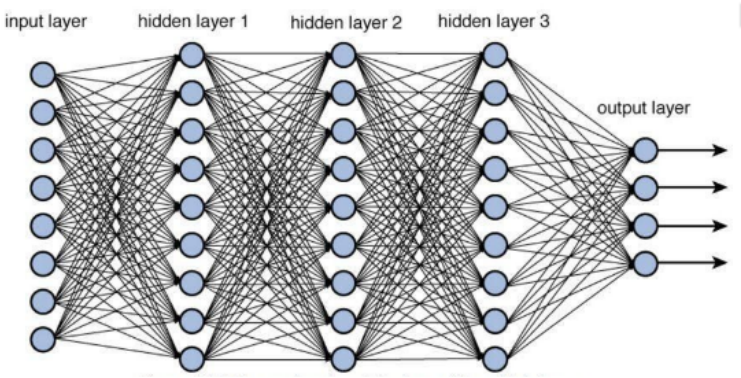


2.4 Deep neural networks

Definition: Deep neural networks

A deep neural network has

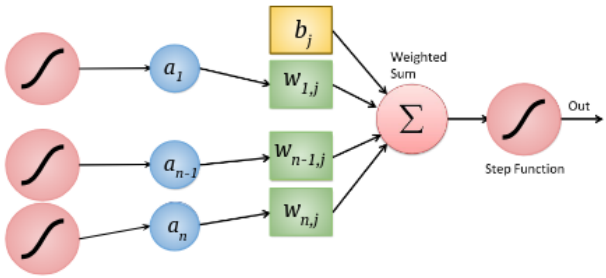
- An input layer: input vector x.
- Hidden layer(s): itermediate representations.
- An output layer: predicting output values for y e.g. probabilities.



In a **fully connected** network, every node in each layer is connected to every node in the next layer. A unit calculates the weighted sum of the inputs from predecessor nodes and plies some activation function to it. Let a_j denote the output of unit j, let $w_{i,j}$ be the weight attaching unit i to unit j. Let b_j be the bias and g_j the activation function of unit j. Then

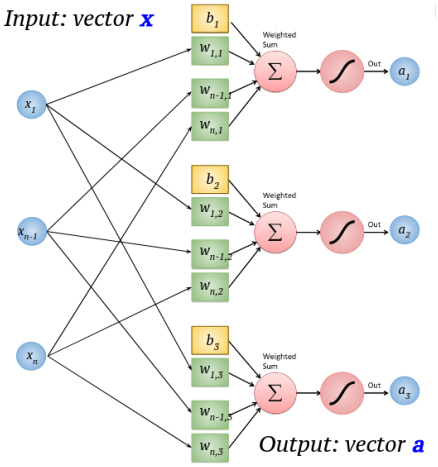
$$\begin{aligned} a_j &= g_j\Big(\sum_i w_{i,j}a_i + b_j\Big) \\ &= g_j(\mathbf{w}_j^\top \mathbf{x} + b_j) \end{aligned}$$

where $\mathbf{w}_j$ is the vector of weights into unit j and $\mathbf{x}$ is the vector of inputs for unit i.



This describes the operation of a single unit in vector form. We can describe entire layers and networks in matrix form.

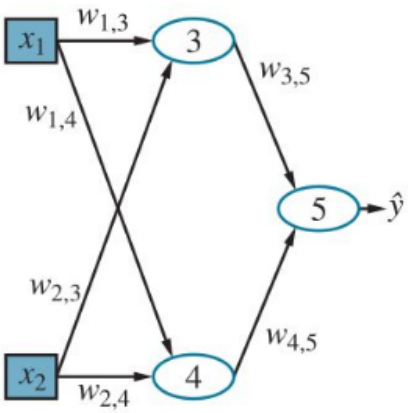
2.4.1 Matrix notation



Each unit a weight vector $\mathbf{w}_i$ and a bias term b_i. Combine the weight vectors into a matrix W, and the biases into a vector $\mathbf{b}$:

$$W = \begin{bmatrix} \leftarrow w_1^\top \rightarrow \\ \leftarrow w_2^\top \rightarrow \\ \leftarrow w_3^\top \rightarrow \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$W^{(i)}$ is the weight matrix of the i-th layer, each row corresponding to the weight vector of a unit. $\mathbf{b}^{(i)}$ is the vector of biases of the i-th layer. Then $\mathbf{a} = g(W\mathbf{x} + \mathbf{b})$.



$$\begin{aligned} \hat{y} &= g_5(\text{in}_5) = g_5(w_{0,5} + w_{3,5}a_3 + w_{4,5}a_4) \\ &= g_5(w_{0,5} + w_{3,5}g_3(\text{in}_3) + w_{4,5}g_4(\text{in}_4)) \\ &= g_5(w_{0,5} + w_{3,5}g_3(w_{0,3} + w_{1,3}x_1 + w_{2,3}x_2) \\ &\quad + w_{4,5}g_4(w_{0,4} + w_{1,4}x_1 + w_{2,4}x_2)) \end{aligned}$$

Here,

- $W^{(1)}$ is the weights of the first layer ($w_{1,3}$, $w_{1,4}$, etc)
- $W^{(2)}$ is the weights of the second layer ($w_{3,5}$, etc)
- $g^{(1)}$ and $g^{(2)}$ are the activation functions in the first and second layer, respectively.

Then the entire network can be written as

$$h_W(\mathbf{x}) = g^{(2)}\big(W^{(2)}g^{(1)}(W^{(1)}\mathbf{x})\big)$$

If we include the biases as well

$$\boxed{h_{W,\mathbf{b}}(\mathbf{x}) = g^{(2)}\big(W^{(2)}g^{(1)}(W^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) + \mathbf{b}^{(2)}\big)}$$

2.5 Cost function

Also known as the loss function. The initialization of weight and bias values are initially random. The purpose of the loss function is to grade the performance of the neural network. High loss signifies large error and poor performance. A low loss indicates that the network's predictions are close to the actual values.

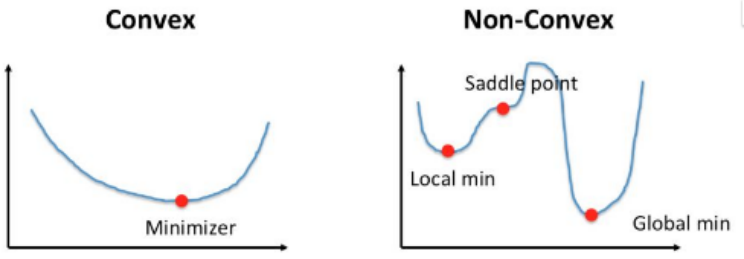
2.6 Optimization

Let the loss function be parameterized by parameters $\theta = (W^{(i)}, \mathbf{b}^{(i)})$ for $i = 1, \ldots, L$ where L is the number of layers. Let's represent our prediction $\hat{y}$ as $f(\mathbf{x}; \theta)$. We want to find weights that minimize the loss, averaged over all examples in our training set $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_m, y_m)\}$

$$\hat{\theta} = \underset{\theta}{\text{argmax}} \frac{1}{m} \sum_{i=1}^{m} L_{\text{CE}}(f(\mathbf{x}^{(i)}; \theta), y^{(i)})$$

There is no closed form solution, so we have to use general purpose techniques for mathemetical optimization. For neural networks:

- The loss function is non-convex (has more than one minimum).
- Gradient descent from any point is guaranteed to find a local minimum, but not necessarily the global minimum.
- Gradient descent still works in practice. It finds a minima that leads to good performance.



For n-dimensional gradients:

- The gradient of a function of many variables is a vector pointing in the direction of the greatest increase in a function.
- To determine the step size, the value of the gradient is weighted by a learning rate η.

$$w^{t+1} = w^t - \eta \frac{d}{dw} L(f(x; w), y)$$

The final equation for updating parameters θ based on the gradient is

$$\theta^{t+1} = \theta^t - \eta \nabla_\theta L(f(\mathbf{x}; \theta), y)$$

2.7 Back-propagation

2.7.1 A quick review of the chain rule

The chain rule tells us how to differentiate composite functions. A function is composite if you can write it as $f(g(x))$. In other words, it is a function of a function. The chain rule allows us to write the derivative of the composite function in terms of the derivatives of $g(x)$ and $f(x)$.

$$\boxed{\frac{d}{dx}\big[f(g(x))\big] = f'(g(x)) \cdot g'(x)}$$

We can also write it in Leibniz's notation:

$$f(x) = u(v(x)) \rightarrow \frac{df}{dx} = \frac{du}{dx} \cdot \frac{dv}{dx}$$

$$f(u(v(w(x)))) \rightarrow \frac{df}{dx} = \frac{du}{dx} \cdot \frac{dv}{dx} \cdot \frac{dw}{dx}$$

2.7.2 What is back-propagation?

We have looked at forward-propagation. Now we move to back-propagation. Back-propagation allows a neural network to learn from its mistakes. Its main job is to efficiently calculate how much every signle weight and bias in the network contributed to the final error. A clever and efficient way to solve this problem is to use the chain rule from calculus. A neural network is a series of composite functions. Apply the chain rule to find the derivative of the loss with respect to a specific parameter (e.g the i-th weight of the j-th unit in the k-th layer).

$$\frac{\partial L}{\partial w_i} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial w_i}$$

Derivation of back-propagation

The cost function:

$$J(\hat{y}^{(i)}, y^{(i)}) = \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}(\hat{y}^{(i)}, y^{(i)})$$

The loss function:

$$\mathcal{L}^{(i)}(\hat{y}^{(i)}, y^{(i)}) = -\Big[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)} \log(1 - \hat{y}^{(i)}))\Big]$$

The update rule:

$$w^{[L]} \leftarrow w^{[L]} - \alpha \frac{\partial J}{\partial w^{[L]}}$$

$$b^{[L]} \leftarrow b^{[L]} - \alpha \frac{\partial J}{\partial b^{[L]}}$$

Recall the following derivative rules:

- $f(x) = \log(x)$, then $f'(x) = \frac{1}{x}$.
- $f(x) = \sigma(x)$, then $f'(x) = f(x)(1 - f(x))$.

The derivation:

$$\begin{aligned}
\frac{\partial \mathcal{L}^{(i)}}{\partial w^{[n]}} &= -\Big[y^{(i)} \frac{\partial}{\partial w^{[n]}} \big[\log(\hat{y}^{(i)})\big] + (1 - y^{(i)}) \frac{\partial}{\partial w^{[n]}} \big[\log(1 - \hat{y}^{(i)})\big]\Big] \\
&= -\Big[y^{(i)} \frac{\partial}{\partial w^{n}} \big[\log(\sigma(w^{[n]^\top} a^{[n-1]} + b^{[n]}))\big] + (1 - y^{(i)}) \frac{\partial}{\partial w^{[n]}} \big[\log(1 - \sigma(w^{[n]^\top} a^{[n-1]} + b^{[n]}))\big]\Big] \\
&= -\Big[y^{(i)} \frac{1}{a^{[n]}} \frac{\partial}{\partial w^{n}} \big[\sigma(w^{[n]^\top} a^{[n-1]} + b^{[n]})\big] + (1 - y^{(i)}) \frac{1}{1 - a^{[n]}} \frac{\partial}{\partial w^{[n]}} \big[1 - \sigma(w^{[n]^\top} a^{[n-1]} + b^{[n]})\big]\Big] \\
&= -\Big[y^{(i)} \frac{1}{a^{[n]}} a^{[n]}(1 - a^{[n]}) \frac{\partial}{\partial w^{n}} \big[w^{[n]^\top} a^{[n-1]} + b^{[n]}\big] + (1 - y^{(i)}) \frac{1}{1 - a^{[n]}} (-a^{[n]})(1 - a^{[n]}) \frac{\partial}{\partial w^{[n]}} \big[1 - w^{[n]^\top} a^{[n-1]} + b^{[n]}\big]\Big] \\
&= -\Big[y^{(i)}(1 - a^{[n]})(a^{[n-1]}) - (1 - y^{(i)}) a^{[n]} a^{[n-1]}\Big] \\
&= (a^{[n]} - y^{(i)}) a^{[n-1]^\top}
\end{aligned}$$

Then, the update rule changes to

$$w^{[n]} \leftarrow w^{[n]} - \alpha \frac{1}{m} \sum_{i=1}^{m} (a^{[n]} - y^{(i)}) a^{[n-1]^\top}$$

To progagate back one step further for $L = n - 1$, we get:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w^{[n-1]}} &= \frac{\partial \mathcal{L}}{\partial a^{[n]}} \cdot \frac{\partial a^{[n]}}{\partial z^{[n]}} \cdot \frac{\partial z^{[n]}}{\partial a^{[n-1]}} \cdot \frac{\partial a^{[n-1]}}{\partial z^{[n-1]}} \cdot \frac{\partial z^{[n-1]}}{\partial w^{[n-1]}} \\ &= (a^{[n]} - y^{(i)}) w^{[n]^\top} a^{[n-1]} (1 - a^{[n-1]}) a^{[n-2]^\top} \\ &= w^{[n]^\top} a^{[n-1]} (1 - a^{[n-1]}) (a^{[n]} - y^{(i)}) a^{[n-2]^\top} \end{aligned}$$

... and so on ...

2.7.3 Activation functions

We need activation functions to encapsulate the complexities. If an activation function is not used, then it is equivalent to linear regression. Next, we need to visit the vanishing gradient problem. When does it occur and which activation functions are particularly susceptible to it?

- It occurs where the gradients in the early layers of a deep neual network become extremely small (approaching zero) during training.
- When the gradients are tiny, the updates to the weights in those layers are also minuscule (learning is very slow/stops altogether). Thus, the early layers fail to learn meaningful features from the data.
- The susceptible functions are **sigmoid** and **hyperbolic tangent**.

3 Convolution Neural Networks

3.1 Computer vision

3.1.1 How computers "see" images?

While humans perceive images as a continuous stream of light, computers are fundamentally discrete and interpret images as numerical matrices of pixel values. For greyscale images, we have a $H \times W$ matrix where each pixel value typically ranges from 0 to 255. For color images (RGB), we have a $H \times W \times 3$ tensor where each pixel contains three distinct color channels: red, green, and blue.

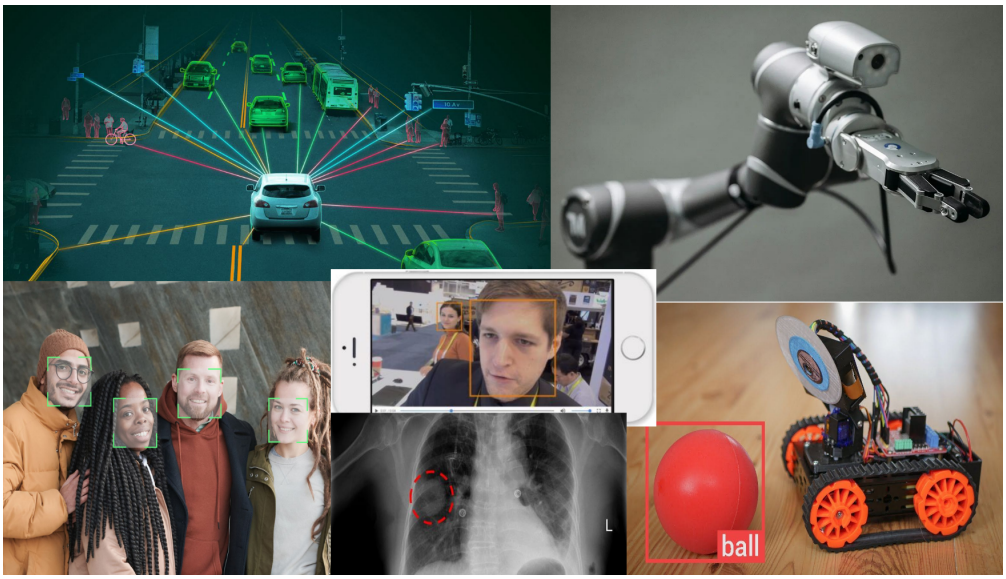
3.1.2 Real-world applications

Computer vision automates visual perception across multiple core tasks, namely:

- image classification,
- object recognition and detection, and
- image segmentation

There are several real-world applications of computer vision:

- Healthcare and medical imaging.
- Autonomous vehicles and transportation.
- Agriculture (AgriTech).
- Retail and customer experience.
- Security and Surveillance.
- Environmental Monitoring and Conservation.



3.2 Why do MLPs struggle on images?

3.2.1 Spatial adjacency

In an image, pixels that are adjacent are strongly correlated and form meaningful patterns like edges, corners, and textures. When a 2D image is flattened into a 1D array, vertical neighbors are separated across the vector. Standard dense layers treat all input elements identically without encoding geometric or spatial structure.

3.2.2 Parameter explosion

In a dense layer, every input unit connects to every hidden neuron. For a color image of size $1000 \times 1000 \times 3$, the input vector has 3000000 dimensions. Connecting this to a single hidden layer of just 1000 neurons would require $3 \times 10^6 \times 10^3 = 3$ billion weights (plus biases). This computational burden causes expensive memory usage, high latency, and severe overfitting.

3.2.3 Inability to handle visual variations

A standard MLP must re-learn the pattern separately at every single pixel position because it lacks translation invariance. Real-world computer vision must remain invariant to seven fundamental visual challenges:

1. **Intra-class variance:** In other words, each class needs diverse shapes, sizes, colors, etc.
2. **Transformation variance:** Non-rigid transformations. For example, take a dog. When a dog is curled up or lying down, we still need to be able to classify it as a dog.
3. **Viewpoint variance:** Viewing an object from various perspectives and angles.
4. **Scale variance:** Viewing an object appearing very large (close to the camera) or very small (far away from the camera).
5. **Illumination variance:** Shifts in lighting, shadows, and contrast.
6. **Occlusion:** Objects partially hidden behind other objects/entities, i.e. a leopard hiding behind a rock.
7. **Background clutter:** Objects blending into complex environments (camouflage).

3.3 CNN architecture

3.3.1 Local receptive fields

Instead of connecting a hidden neuron to every pixel in the image, each neuron connects only to a small, localized sub-region of the input. This perserves the adjacency of the pixels. This also reduces the weight count significantly. If each region covers k pixels ($k \ll n$), the number of connections per node drops from n to k.

3.3.2 The convolution operation

A **kernel** (or **filter**) is a small learnable weight matrix, typically 2×2. The kernel slides systematically across the input image patch-by-patch. A each location, the network computes an element-wise multiplication between the kernel weights and current image patch, then sums the products (a dot product) to produce a signle value in the feature map.

3.3.3 Multiple kernels per layer

A single filter can only detect one specific feature type. To capture varied visual patterns, each layer uses multiple independent filters. convolving a 6×6 image with two separate 3×3 filters produces two 4×4 feature maps, resulting in an output volume of dimension $4 \times 4 \times 2$.

3.3.4 Feature detection

Although each individual filter remains small (e.g. 3×3), stacking layers allows a neuron in a deeper layer to span a wider area of the original input, giving it a broader context window.

- **Layer 1:** Detects basic visual stimuli such as oriented edges, color transitions, and lines.
- **Layer 2:** Combines edge primitives to detect corners, textures, curves, etc.
- **Layer 3+:** Combines mid-level shapes to recognize complex structural parts and whole objects.

3.4 Convolution operation

3.4.1 2D cross-correlation

For a 2D matrix I and a $k \times k$ kernel K, the activation at output coordinate (i, j) is defined as:

$$S(i,j) = (K * I)(i,j) = \sum_m \sum_n I(i+m, j+n) K(m,n)$$

Example:

$$\underbrace{\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}}_{I} * \underbrace{\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}}_{K} + \underbrace{5}_{b} = \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix}$$